

NGUYEN TUAN KIET

BACKEND DEVELOPER

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OBJECTIVE

Backend Developer (Java Spring Boot, Node.js) with hands-on experience building RESTful APIs, microservices and payment integrations. AI-native in daily work: using Claude Code, Copilot and AI agents to ship faster, while carefully reviewing AI-generated code for quality and security. Seeking a Fresher/Junior Backend role to build reliable, AI-powered products and keep up with fast-changing technology.

EDUCATION

2020 - 2025	**Graduated from FPT University Da Nang** Software Engineering

WORK EXPERIENCE

11/2025 - 03/2026	**Startup Company – Construction Technology** **Backend Developer Intern (Node.js)** *Internal management system for a construction-tech company.* • **API development:** Built RESTful APIs with Node.js (Express.js) for the internal management system. • **Database design:** Designed MongoDB collections and data structures supporting the system's core features. • **Teamwork:** Collaborated through a Git workflow and resolved merge conflicts during development. **Tech:** Node.js, Express.js, MongoDB, RESTful API, Git
05/2024 - 09/2024	**FPT Software** **Web Developer Intern (Java Spring Boot & ReactJS)** *Full-stack web application development in an Agile/Scrum team.* • **Full-stack features:** Developed web features with Spring Boot (backend) and ReactJS (frontend). • **API integration:** Built RESTful APIs and connected them to the frontend. • **Business logic:** Implemented backend business logic following the MVC architecture. • **Quality:** Tested, debugged, maintained and enhanced existing features. **Tech:** Java, Spring Boot, Spring MVC, ReactJS, RESTful API, Git, Agile/Scrum
04/2026 - 09/2026	**VNPT** **Network Infrastructure Technician** *Field deployment of network infrastructure for household and business customers.* • **Deployment:** Installed and deployed network cabling infrastructure at customer sites to company technical standards. • **Connectivity:** Tested connections after installation to ensure stable service handover. • **Soft skills:** Worked directly with customers and field teams; handled on-site tasks to schedule and safety requirements. **Skills:** Network cabling, LAN fundamentals, connection testing, field work, customer communication

PROJECTS

11/2026

Seat Booking API – No Double-Selling
Personal Project
Source: github.com/KietNguyen-Java/seat-booking-api
Event ticketing API that never sells a seat twice, even when many customers book at the same moment.

- **Concurrency:** Row locks with SELECT ... FOR UPDATE in id order to prevent deadlocks; a 30-thread test on one seat yields exactly one hold.
- **Idempotency:** Idempotency-Key plus a SHA-256 of the request body, so client retries never create duplicate holds.
- **Hold expiry:** Scheduled job with FOR UPDATE SKIP LOCKED releases seats unpaid after 5 minutes.
- **Best seats:** O(n) sliding-window algorithm picks k adjacent seats closest to the stage and the row centre.

Tech: Java 21, Spring Boot, JPA/Hibernate, PostgreSQL, Flyway, JUnit 5, GitHub Actions (CI)

11/2026

Da Nang Route Finder – Routing API
Personal Project
Source: github.com/KietNguyen-Java/danang-route-finder
REST API for routing on a map of 24 places and 36 roads in Da Nang, stored in PostgreSQL.

- **Graph algorithms:** Implemented Dijkstra, A* (haversine heuristic), BFS and Yen's k-shortest paths; verified against Bellman-Ford and brute force.
- **Tour planning:** Solves the travelling salesman problem exactly with Held-Karp dynamic programming, $O(2^n \cdot n^2)$, for up to 12 stops.
- **Database:** Flyway-managed schema, CHECK/UNIQUE constraints, a (latitude, longitude) index for nearby search, confirmed with EXPLAIN.
- **Performance:** Immutable in-memory graph cache refreshed after commit; CI runs the tests on H2 and PostgreSQL.

Tech: Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, JUnit 5, GitHub Actions (CI)

11/2026

MiniDB – A Database Engine from Scratch
Personal Project
Source: github.com/KietNguyen-Java/minidb
A small SQL database written in plain Java with no dependencies outside the JDK.

- **SQL parser:** Hand-written lexer and recursive-descent parser for CREATE, INSERT, SELECT, UPDATE, DELETE, JOIN, ORDER BY, LIMIT and EXPLAIN.
- **B+ tree indexes:** Full B+ tree (split, borrow, merge); property-tested with 20,000 random operations against TreeMap.
- **Query planner:** Chooses index lookup, range scan or full scan, and hash, index nested-loop or nested-loop join.
- **Durability:** Write-ahead log with CRC32 and recovery from torn writes; every statement is atomic via an undo log.

Tech: Java 21, B+ tree, SQL parser, Write-ahead log, JUnit 5, GitHub Actions (CI)

07/2026 - 09/2026

Online Printing Platform for Students (Freelance)
Freelance Developer
Web platform that lets students order document printing online and pay directly on the site.

- **Ordering flow:** Built the flow for students to upload documents, choose print options and place orders online.
- **Payments:** Integrated online payment so orders are paid and confirmed automatically.
- **Hardware integration:** Connected the web system with printer hardware to process print jobs from online orders.
- **Delivery:** Worked directly with the client from requirements to deployment as a freelancer.

Tech: Java, Nginx, PayOS API, GitHub Actions (CI/CD)

12/2025 - 12/2025	**Landing Pages for Clients** **Frontend Developer** *Landing pages built from client-provided designs.* • **Design to code:** Converted client designs into responsive landing pages faithful to the original layout. • **AI-assisted animation:** Used AI coding tools to implement the required animations and interactions quickly. • **Client delivery:** Iterated on client feedback and delivered pages on schedule.
05/2025 - 08/2025	**CUSTOMERS – AI-integrated Real Estate Platform** **Backend Developer** *Real estate sales & management platform built on a microservices architecture.* • **Authentication:** Built registration/login with JWT, bcrypt, email OTP verification and user role authorization. • **Payments:** Integrated PayOS API to process broker commission payments, improving transaction transparency. • **Core services:** Developed broker review system with approval flow, appointment management and blog management. • **Notifications:** Automated email notifications (Nodemailer) for multiple business workflows. • **Architecture & DevOps:** Built independent microservices behind Nginx; set up CI/CD with GitHub Actions. **Tech:** Node.js, Express.js, Prisma ORM, MySQL, JWT, bcrypt, Microservices, Nginx, PayOS API, Nodemailer, GitHub Actions (CI/CD)

CERTIFICATIONS

22/03/2026	Certification Claude Code – AI-Assisted Software Development, Skilljar | 2026 https://verify.skilljar.com/c/3bhr4cwchks6
03/11/2024	Project Management Certification, Coursera https://www.coursera.org/account/accomplishments/certificate/R1GMCG6WEJU5
03/11/2024	Project Change and Risk Management Certificate, Coursera https://www.coursera.org/account/accomplishments/certificate/TKG1SXO57M69
03/11/2024	Budgeting and Scheduling Projects, Coursera https://www.coursera.org/account/accomplishments/certificate/3N4PY0EVZSNO
17/09/2024	Initiating and Planning Projects, Coursera https://www.coursera.org/account/accomplishments/certificate/0WYG5294A0CB

SKILLS

Programming languages:	Java, JavaScript
Framework:	Spring Framework, Spring MVC, Spring IoC, Spring Boot, Spring Data, RESTful API, MVC Architecture
Backend Development:	Full-stack Application Development, Backend Development, Web Application Development, Microservices, REST Web Services
Algorithms & Data Structures:	Graphs (Dijkstra, A*, BFS, Yen's k-shortest paths), dynamic programming (Held-Karp), B+ trees, hash tables, sliding window; property tests against brute force
ORM & Database:	PostgreSQL, MySQL, MongoDB, JPA/Hibernate, Prisma ORM, Flyway; database and index design, transactions and locking (SELECT FOR UPDATE, optimistic locking), reading EXPLAIN, query optimization
Testing:	Unit Testing, Integration Testing, Concurrency Testing, Software Testing, Code Quality, Debugging

Frontend Development:	ReactJS
Tools:	Git, Docker, Maven, GitHub Actions (CI/CD)
Office software:	Microsoft Word, Excel, PowerPoint.
Web Services & API Integration:	RESTful Web Services, REST API Development, Third-party API Integration, Payment API Integration, Idempotency
Authentication & Security:	JWT, Authentication, Authorization, OTP Authentication, bcrypt, Web Application Security
Software Development Processes:	Agile/Scrum, Design Patterns, Technical Documentation, Software Development Lifecycle, Code Maintenance, Application Enhancement
AI Coding Tools:	Claude Code (certified), GitHub Copilot, Cursor, ChatGPT
AI-assisted Workflow:	Prompt engineering for AI coding agents (Openclaw), AI-assisted code review & refactoring, unit test generation, debugging & log analysis, API documentation generation
AI Integration & Responsible Use:	Integrating LLM APIs (OpenAI, Claude) into backend services, structured JSON output, basic RAG & vector databases; reviewing and validating AI-generated code for correctness and security
Professional skills:	Strong Analytical Skills, Problem Solving, Quick Comprehension of Existing Applications, Ability to Quickly Learn and Apply New Technologies

INTERESTS

- **AI:** Researching and following new AI models, tools and coding agents; trying them in personal projects.
- **Self-hosting:** Building home servers, setting up home/LAN networks and deploying my own web projects.
- **Single-board computers:** Tinkering with Raspberry Pi 4/5 and similar devices for self-hosted services and small hardware projects.

Others: soccer, guitar, traveling.

ADDITIONAL INFORMATION

I am well aware that programming work requires meticulousness, the ability to optimize source code and meet strict deadlines. For me, this is an opportunity to challenge myself, not an obstacle, thanks to the spirit of continuous learning.